

Event Registration System Deployment on Microsoft Azure

Project Name

Event Registration System

GitHub Repository

[EventRegistration-System GitHub Repository](#)

Aim

To deploy the Event Registration System project on a Microsoft Azure Virtual Machine using:

- Ubuntu Linux VM
- Node.js
- MongoDB Atlas
- PM2 Process Manager
- Nginx Web Server

Technologies Used

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB Atlas
- Server: NGINX
- Process Manager: PM2
- Cloud Platform: Microsoft Azure
- Version Control: Git & GitHub

Azure VM Configuration

- OS: Ubuntu 24.04 LTS
- VM Size: Standard B1s
- Authentication Type: SSH Key Pair

Deployment Steps

Step 1: Connect to Azure VM

```
cd Downloads
```

```
ssh -i VM1_key.pem azureuser@<PUBLIC-IP>
```

Step 2: Update Ubuntu Packages

```
sudo apt update && sudo apt upgrade -y
```

Step 3: Install Node.js

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E  
bash -
```

```
sudo apt install -y nodejs
```

Check Version

```
node -v  
npm -v
```

Step 4: Install Git and Nginx

```
sudo apt install git nginx -y
```

Step 5: Clone GitHub Repository

```
cd ~
```

```
git clone https://github.com/GauriWagh1109/EventRegistration-  
System.git
```

Open Project Folder

```
cd EventRegistration-System
```

Backend Deployment

Step 6: Install Backend Dependencies

```
npm install
```

Step 7: Create Environment Variables

Create `.env` file:

```
nano .env
```

Add:

```
PORT=4000
```

```
MONGO_URI=your_mongodb_connection_string
```

Step 8: Install PM2

```
sudo npm install -g pm2
```

Step 9: Start Backend Server

```
pm2 start server.js --name backend
```

Check Running Process

```
pm2 list
```

Save PM2 Process

```
pm2 save
```

Step 10: Test Backend

```
curl http://localhost:4000/registrations
```

Sample Output

```
[
  {
    "_id": "12345",
```

```
    "name": "Gauri",
    "email": "gauri@gmail.com",
    "college": "ABC College",
    "event": "Coding Competition"
  }
]
```

Frontend Deployment

Step 11: Build React Frontend

```
cd ~/EventRegistration-System/frontend
```

```
npm install
```

```
npm run build
```

Step 12: Configure Nginx

Create Nginx Configuration:

```
sudo nano /etc/nginx/sites-available/app
```

Add Configuration:

```
server {
    listen 80 default_server;
    server_name _;

    root /home/s24ce016/EventRegistration-System/frontend/
build;
    index index.html;

    location / {
        try_files $uri /index.html;
    }

    location /registrations {
        proxy_pass http://localhost:4000;
    }
}
```

Step 13: Enable Nginx Configuration

```
sudo ln -s /etc/nginx/sites-available/app /etc/nginx/sites-enabled/
```

Remove Default Configuration

```
sudo rm /etc/nginx/sites-enabled/default
```

Step 14: Fix Folder Permissions

If you get permission denied or 500 Internal Server Error:

```
sudo chmod 755 /home/s24ce016
```

```
sudo chmod -R 755 /home/s24ce016/EventRegistration-System
```

```
sudo chmod -R 755 /home/s24ce016/EventRegistration-System/
frontend/build
```

Step 15: Test and Restart Nginx

```
sudo nginx -t
```

```
sudo systemctl restart nginx
```

Step 16: Allow HTTP Port in Azure

Go to Azure VM Networking Settings:

- Add Inbound Rule
- Port: 80
- Protocol: HTTP
- Action: Allow

Errors Faced During Deployment

1. 500 Internal Server Error

Cause

Incorrect Nginx root path and permission issues.

Solution

Updated Nginx configuration with correct React build folder path:

```
root /home/s24ce016/EventRegistration-System/frontend/build;
```

Also fixed folder permissions using chmod.

2. Permission Denied Error

Cause

Nginx user (www-data) could not access files inside /home/s24ce016.

Error Log

```
Permission denied
```

Solution

```
sudo chmod 755 /home/s24ce016
```

```
sudo chmod -R 755 /home/s24ce016/EventRegistration-System
```

3. Rewrite or Internal Redirection Cycle Error

Cause

Incorrect React routing configuration in Nginx.

Error

```
rewrite or internal redirection cycle while internally  
redirecting to "/index.html"
```

Solution

Updated routing configuration:

```
location / {  
    try_files $uri /index.html;  
}
```

4. Backend Connection Error

Cause

Incorrect backend port used in Nginx configuration.

Solution

Backend was running on port 4000, so updated:

```
proxy_pass http://localhost:4000;
```

Final Output

<http://20.244.14.63/>

The Event Registration System was successfully deployed on Microsoft Azure using Ubuntu VM, PM2, MongoDB Atlas, and Nginx.